TRIMESTER 1 PROJECT

Credit Risk & Fraud Analytics

Exploratory data analysis, a first credit-risk model, and a practical monitoring view for a fictional digital lender.

Prepared for

Credit Risk Committee | **Lending Operations** | **Fraud Monitoring** | **Data & IT**

20,000	**24,831**	**120,000**	**50,000**
Current loan applications	Previous applications	Instalment records	Transactions
15.82%	**5.3:1**	**1.74%**	**0.786**
Payment difficulty rate	Majority / minority class	Simulated fraud-label rate	Baseline ROC-AUC

Data note: All records, outcome labels and reported results in this project are synthetic. This report is for learning and discussion only. It does not describe real bank performance and must not be used to make actual lending or fraud decisions.

Project scope

Python data generation, data quality checks, EDA, conditional probabilities, a Welch t-test, within-segment correlations, a logistic-regression baseline, an expected-loss example, transaction-fraud pattern checks, and practical BI/IT recommendations.

1. Executive summary

The aim is to understand which application-time fields are associated with payment difficulty, show how a basic model ranks risk, and identify transaction patterns that a monitoring team could review. A second goal is to practise an end-to-end workflow that creates its own data and produces repeatable analysis.

What stands out in this synthetic sample

- 15.8% of current loans are labelled as having payment difficulty. The class ratio is about 5.3:1, so accuracy alone is not enough to assess a model.
- Default rates rise across lower credit-score bands and higher debt-to-income (DTI) bands. This is a simulated relationship built into the data for practice.
- The average credit score is 642.9 for payment-difficulty cases and 679.4 for other cases. A Welch t-test finds a clear difference in this sample (p-value 1e-210); this does not prove cause and effect.
- The baseline logistic model has ROC-AUC 0.786, but recall at the default 0.50 cut-off is only 18.5%. It misses many payment-difficulty cases, so this is not a deployment-ready decision rule.
- 1.74% of synthetic transactions carry the fraud label. New-device transactions show a 4.34% label rate, compared with 1.59% for usual devices.

Dataset inventory

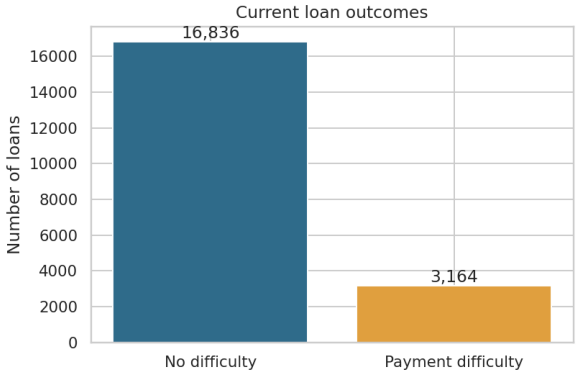
File	Rows	Purpose
application_data.csv	20,000	Current loan features and simulated repayment outcome
previous_application.csv	24,831	Historical applications, decisions and past-due days
repayment_history.csv	120,000	Six post-loan instalment records per applicant
transactions.csv	50,000	Transaction attributes and simulated fraud label
data_dictionary.csv	14 fields described	Short definitions and notes

Analysis flow

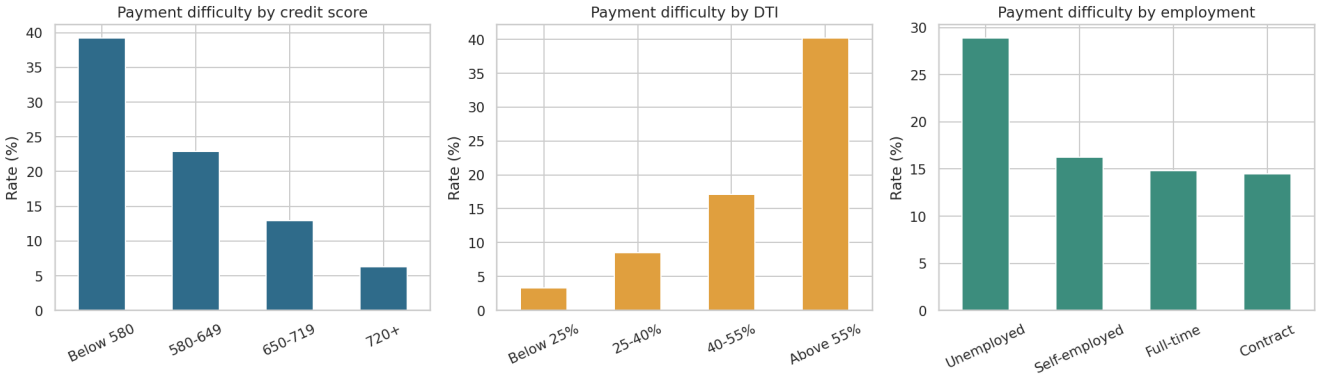
Generate data -> check shape, duplicates and missing values -> create business measures -> compare outcomes -> inspect correlations and probability -> train a baseline model -> translate results into risk, operations and IT actions.

2. Credit-risk EDA

The charts focus on rates rather than only counts because the two target classes have different sizes. Risk patterns are interpreted as signals for further review, not as proof that a feature causes default.



Payment difficulty by risk segments



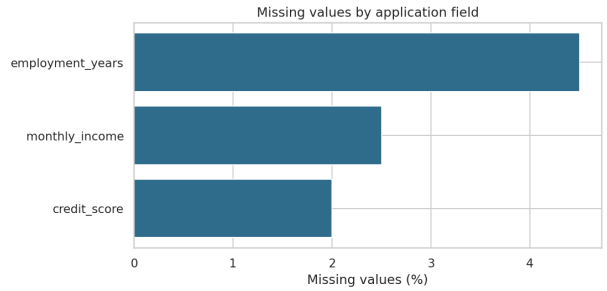
Selected rates from the generated sample

Segment	Observed rate	Reading
Credit score: Below 580	39.2%	Higher at lower scores
Credit score: 580-649	22.8%	Compare against other bands
Credit score: 650-719	13.0%	Compare against other bands
Credit score: 720+	6.3%	Compare against other bands
DTI: Below 25%	3.3%	Compare with income and loan terms
DTI: 25-40%	8.5%	Compare with income and loan terms
DTI: 40-55%	17.1%	Compare with income and loan terms
DTI: Above 55%	40.2%	Higher debt burden is linked with more difficulty

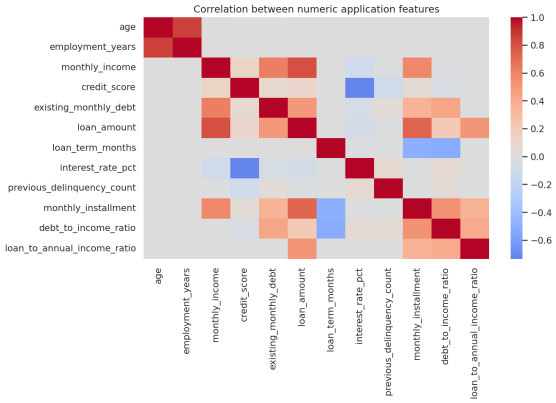
Business reading: a lower credit score, higher repayment burden and previous late payments are useful fields to review together. Any real policy would need validation against actual outcomes, affordability rules, customer fairness and operational capacity.

3. Data quality and statistical checks

Missing values and outliers are reported before modelling. The IQR method is used to flag unusual numeric values, not to delete them automatically.



Missing values were inserted only in a few fields for cleaning practice. The model fills numeric values using medians from its training set.



The strongest pairs include age with employment years, income with loan amount, and credit score with offered interest rate. Correlation does not show cause.

Quality check summary

Field	Missing share	Handling / note
Employment years	4.5%	Keep visible in EDA; median-fill for the baseline model
Monthly income	2.5%	Keep missingness visible; training-set median used for modelling
Credit score	2.0%	Keep missingness visible; training-set median used for modelling
Potential outliers	IQR method	Count and investigate; do not remove without business checks

Probability and correlation points

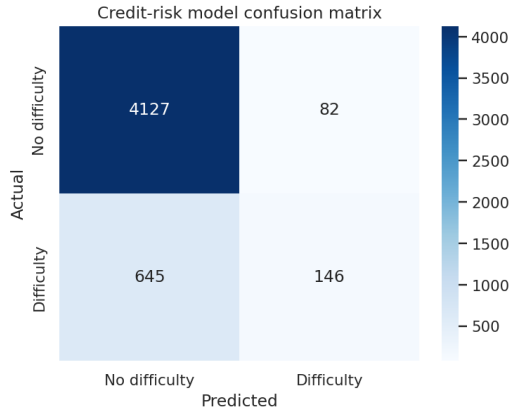
- P(payment difficulty) = 0.158 in the synthetic application sample.
- P(payment difficulty | any previous 30+ day delinquency) = 0.261, compared with 0.152 when no previous delinquency was recorded.
- The Welch t-test compares mean credit scores by target group: t = -32.78, p = 1e-210. Because the dataset is synthetic, this result only checks that the designed pattern is detectable.

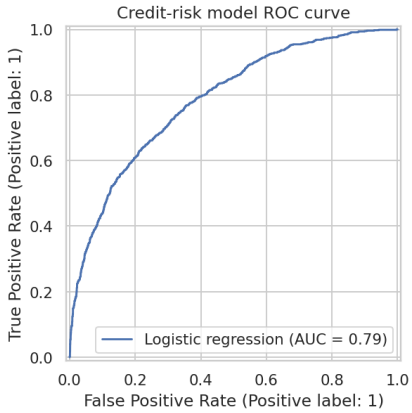
The notebook also prints the top 10 numeric feature-pair correlations separately for each target segment, with the target excluded from these pair tables as required by the assignment brief.

4. Baseline model and expected-loss example

A logistic-regression baseline uses eight numeric fields available at application time. The test set contains 5,000 applicants. Data is split before imputation and scaling; this keeps test data out of the fitting steps.

Metric	Test result	Why it matters
Accuracy	0.855	Overall share correctly classified; can hide missed minority cases
Precision	0.640	Share of flagged applicants who actually have the target label
Recall	0.185	Share of payment-difficulty cases found by the model
F1 score	0.287	Balances precision and recall
ROC-AUC	0.786	Ability to rank positive cases above negative cases across cut-offs





At the default 0.50 cut-off, the model catches only a small share of payment-difficulty cases. The next step would be to compare thresholds based on review capacity and the relative cost of missed defaults and unnecessary flags - not to use the current output automatically.

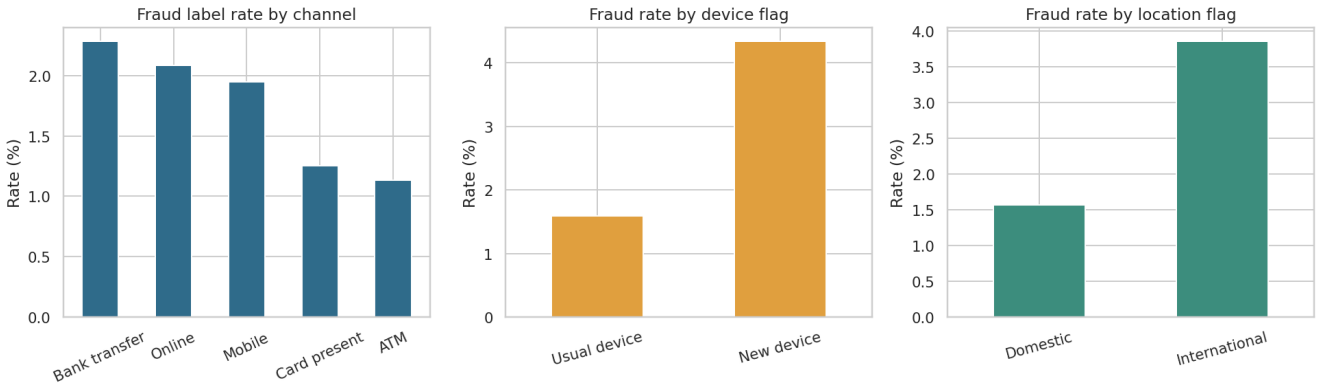
Expected-loss illustration

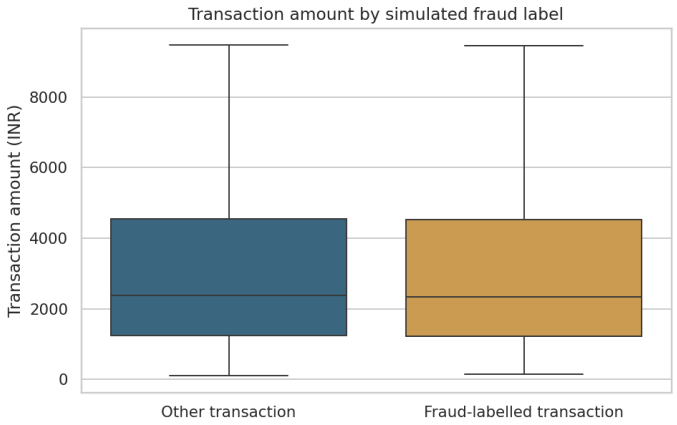
Measure	Value
Assumed loss given default (LGD)	45% of exposure
Test-set applications	5,000
Manual-review share at 20% model score	27.5%
Observed difficulty rate among flagged cases	35.7%
Mean estimated loss per test application	INR 25,348
Total estimated loss across test set	INR 126,741,429

Expected loss = predicted probability of default x loan exposure x assumed LGD. The 45% LGD and 20% manual-review line are learning assumptions, not approved policy inputs.

5. Fraud monitoring and operations

The transaction data is separate from the credit-risk application data. The fraud chart shows which groups carry a higher simulated fraud-label rate. It does not show that every transaction in a group is fraudulent.





Signal / group	Fraud-label rate
Overall transactions	1.74%
New device	4.34%
Usual device	1.59%
International	3.86%
Domestic	1.57%

Operations notes

- Historical application statuses (Approved, Cancelled, Refused and Unused offer) should be reviewed by count and by 30+ day past-due rate, not by count alone.
- Post-loan instalment records help with portfolio monitoring, collections and early-warning dashboards. They must not be used in an application-time model.
- Fraud review can combine channel, international status, new-device flag, unusual hour and amount. A single flag should trigger review rather than an automatic block.

6. BI / IT recommendations and next steps

The practical output is a small risk-monitoring workflow: a repeatable data refresh, a set of understandable KPIs, and a review queue for unusual cases. The first version should favour explainability over complexity.

Suggested BI view

View	Core measures	Who uses it
Credit portfolio	Default rate, target imbalance, default by score/DTI band, average exposure, estimated loss	Credit Risk / Finance
Loan operations	Previous decision counts, delinquency rate by previous status, payment status by month	Lending Operations / Collections
Fraud monitoring	Fraud-label rate by channel/device/location, high-amount transactions, daily trends	Fraud Operations
Data quality	Missing fields, duplicate IDs, outlier counts, input row counts, failed validation checks	Data / IT

Suggested IT workflow

- **Ingest:** store dated CSV extracts in a controlled folder or database table, with a refresh timestamp and row counts.
- **Validate:** check unique keys, target values, expected columns, missing rates and date formats before analysis.
- **Transform:** create DTI and loan-to-income measures in a documented step; keep a clear distinction between application-time and post-loan fields.
- **Monitor:** publish a small KPI summary and alert when missing rates, default rates or transaction flags move outside agreed ranges.
- **Govern:** restrict access to sensitive data, log model versions, review false positives/negatives, check customer fairness, and keep a human decision-maker accountable.

Recommendations for this learning project

- Keep credit score, repayment burden and past delinquency as candidate risk signals, but validate them on real, representative, time-separated data before any use.
- Evaluate a range of decision thresholds and show how recall, precision and the number of manual reviews change.
- For fraud, compare signals in combination and track false alarms. Do not equate a suspicious pattern with confirmed fraud.
- Next trimester, extend only after the foundations are understood: validation by time period, more careful model comparison, threshold analysis, and drift monitoring.

Limitations: Synthetic data, simulated outcomes, simplified LGD, no real collections/recovery data, no credit-bureau records, no fairness assessment, no regulatory validation and no deployment. The project is a practice template, not a live credit or fraud system.